

Shabab Ahmed

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PROFESSIONAL PROFILE

Ph.D. Candidate in Economics specializing in contract theory and structural modeling. Applied the latter during a pricing-focused economics internship at Amazon.

EDUCATION

Ph.D. in Economics, University of Washington

Expected December 2026

Dissertation:

- [Dynamic Moral Hazard under Preference Change](#): how anticipated changes in future preferences affect incentives today, applied to the NFL draft policy. Currently testing the model's predictions using quasi-experimental methods.
- [High-Dimensional Demand Estimation](#): applies Lasso and Elastic Net regularization with debiasing corrections to estimate elasticity under collinearity with potentially many instruments and regressors. Validated via Monte Carlo simulation. Currently extending this method to merger simulation, testing whether predicted price effects are sensitive to the choice of demand specification.
- [Fixed Budget Allocation under Hidden Effort](#): how agents spend across time under a budget constraint. This offers a new explanation for observed patterns like budget hoarding and end-of-year spending spikes.

Passed general examination; presented research in departmental seminars.

M.A. in Economics, University of Washington

Dec 2022 — GPA 3.89

M.S. in Applied Mathematics, University of Washington

Jun 2020 — GPA 3.93

B.A. in Mathematics and Economics, Colby College

May 2019 — GPA 3.93

Magna Cum Laude, Phi Beta Kappa, Distinction in Economics & Mathematics, Dean's List (6 semesters).

EXPERIENCE

Economist Intern, Amazon, Bellevue, WA

Jul 2025 – Oct 2025

- Built structural demand models on large-scale survey data to estimate customer preferences for Project Kuiper. Authored and presented a technical and business report to senior leadership across multiple teams, who evaluated it as an input to pricing decisions.
- Built a Python tool that returned projected market share and revenue based on user-entered pricing and product configurations. The team used it to run counterfactual business scenarios.

Graduate Teaching Assistant & Independent Instructor, University of Washington

2019 – Present

- Independent Instructor for Game Theory and Managerial Economics, with full responsibility for course delivery, assignments, and assessment.
- Teaching Assistant for Graduate Microeconomic Theory and undergraduate courses, leading discussion sections and review sessions.
- Summer Study Section Instructor, Math & Science Upward Bound (Summers 2024 and 2026): mentored high school students on independent research projects.

EARLIER PROJECTS

[NBA Decade Comparison: PCA and Machine Learning Analysis](#)

Winter 2020

- Partnered with a classmate to build classifiers on 13,000 player-season observations (~95 features scraped from Basketball Reference). Used PCA and LDA to predict player position and decade. Used L1-regularized logistic regression to predict Win Shares quartile.
- Achieved 72.6% accuracy predicting Win Shares quartile for 1990s-decade players, cross-validated over 50 iterations.

[Game Theory in Cricket](#)

Spring 2019

- Built a game-theoretic model of bowler-batsman strategy and tested Nash equilibrium predictions against a hand-coded dataset of bowler-batsman interactions from 2017 ICC Champions Trophy match commentary.
- Found predicted strategy frequencies matched actual play within one bootstrapped standard deviation under a simplified model, though the result did not hold under more granular or alternate specifications.

TECHNICAL SKILLS

Causal Inference & Econometrics: IV, DiD/event study

Structural & Statistical Methods: structural demand estimation, dynamic optimization, linear and logistic regression, Lasso/Elastic Net regularization, debiased ML, PCA, LDA, bootstrapping, cross-validation, Monte Carlo simulation

Languages & Tools: Python, R, Stata, SQL, MATLAB, Sage, Mathematica, LaTeX, Qualtrics, Excel

Languages: English (fluent), Bengali (fluent), Hindi (conversational)